Algorithms and Data Structures

REPORT LAB 12

Name: ILGIN
Surname: CEYLAN
Student ID: 82862

TASK 1

```
-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=M
t-MIEngine-Out-j2ucf3ls.adq' '--stderr=Microsoft-MIEngine-E
13yk.lul' '--dbgExe=C:\TDM-GCC-64\bin\gdb.exe' '--interprete
Nodes Finish Times:
Node 0: 10
Node 1: 7
Node 2: 9
Node 3: 5
Node 4: 18
Node 5: 6
Node 6: 17
Node 7: 16
Node 8: 15

Strongly Connected Components:
Component 1: 4 6 7 8
Component 2: 0 1 5 3 2
PS C:\Users\ILGIN\OneDrive\Desktop\lab_07>
```

TASK 2

```
PS C:\Users\ILGIN\OneDrive\Desktop\lab_07>  & 'c:\Users\ILGIN\.vscode\extensions\ms-vscode.cpptools-1.32.2-win32
-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-lhzgws2x.32v' '--stdout=Microsof
t-MIEngine-Out-czbvmiub.ad5' '--stderr=Microsoft-MIEngine-Error-ckvuwjxo.5af' '--pid=Microsoft-MIEngine-Pid-2ink
ujsw.q43' '--dbgExe=C:\TDM-GCC-64\bin\gdb.exe' '--interpreter=mi'
--- Prim's Algorithm Steps ---
Current Min-Heap: [ (Node:0, Key:0) (Node:1, Key:2147483647) (Node:2, Key:2147483647) (Node:3, Key:2147483647) (
Node:4, Key:2147483647) (Node:5, Key:2147483647) (Node:6, Key:2147483647) (Node:7, Key:2147483647) (Node:8, Key:
2147483647) ]
Current Min-Heap: [ (Node:1, Key:4) (Node:7, Key:8) (Node:2, Key:2147483647) (Node:8, Key:2147483647) (Node:4, K
ey:2147483647) (Node:5, Key:2147483647) (Node:6, Key:2147483647) (Node:3, Key:2147483647) ]
Current Min-Heap: [ (Node:7, Key:8) (Node:3, Key:2147483647) (Node:2, Key:8) (Node:8, Key:2147483647) (Node:4, K
ey:2147483647) (Node:5, Key:2147483647) (Node:6, Key:2147483647) ]
Current Min-Heap: [ (Node:6, Key:1) (Node:8, Key:7) (Node:2, Key:8) (Node:3, Key:2147483647) (Node:4, Key:214748
3647) (Node:5, Key:2147483647) ]
Current Min-Heap: [ (Node:5, Key:2) (Node:8, Key:6) (Node:2, Key:8) (Node:3, Key:2147483647) (Node:4, Key:214748
3647) ]
Current Min-Heap: [ (Node:2, Key:4) (Node:4, Key:10) (Node:8, Key:6) (Node:3, Key:14) ]
Current Min-Heap: [ (Node:8, Key:2) (Node:4, Key:10) (Node:3, Key:7) ]
Current Min-Heap: [ (Node:3, Key:7) (Node:4, Key:10) ]
Current Min-Heap: [ (Node:4, Key:9) ]

--- Minimum Spanning Tree (Final MST) ---
Edge    Weight
0 - 1   4
5 - 2   4
2 - 3   7
3 - 4   9
6 - 5   2
7 - 6   1
0 - 7   8
2 - 8   2
Total MST Weight: 37
PS C:\Users\ILGIN\OneDrive\Desktop\lab_07>
```